

Information Gap Decision Theory-Based Active Distribution System Planning for Resilience Enhancement

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Abstract—Severe disruptions caused by natural disasters have highlighted the significance of enhancing the distribution system resilience in recent years. However, due to severe uncertain characteristics of natural disasters, there exists an insufficient level of information to evaluate their behavior and consequences. Lack of information such as proper distribution functions for probabilistic methods or known intervals for robust optimization techniques may provide Distribution System Planners (DSPs) with inaccurate designs. This paper presents a new strategy based on Information Gap Decision Theory (IGDT) to facilitate decision making for DSPs in the face of extreme events. Moreover, a detailed analysis is performed to find a trade-off between the DG allocation and the hardening schemes, and to determine the best conservativeness degree in the IGDT approach. The presented resilience model is implemented on 33-bus and 94-bus distribution systems, and numerical case studies are analyzed to demonstrate the merits of the proposed approach. The results indicate that with a limited planning budget, the proposed trade-off analysis can optimally find the best DG allocation and hardening schemes to keep the load shedding level below a desired threshold following a severely uncertain extreme event.

Index Terms—Resilience, extreme events, distribution system planning, risk-averse strategy.

NOMENCLATURE

Sets and Indices

B	Set of distribution system buses.
L	Set of distribution system lines.
i	Index for buses.
i_s	Index for substation buses.
(i, j)	Index for lines.
k	Index for bus commodities.
$\mathbb{R}$	Set of real numbers.

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$\mathbb{R}_+$ Set of non-negative real numbers.

Parameters

R_{ij}, X_{ij}	Resistance and reactance of distribution line ij .
V_0	Reference voltage magnitude.
$V_i^{max, min}$	Upper and lower bound of bus voltages.
P_i^D, Q_i^D	Active and reactive power of loads at bus i .
S_{ij}^{max}	Maximum permissible capacity of line ij .
$P_{ij}^{G, max}$	Maximum active power of DGs.
$Q_{ij}^{G, max, min}$	Maximum and minimum reactive power of DGs.
$C_{i,j}$	Candidate lines in distribution network.
$\ell_{i,j}$	Length of distribution lines.
ω_i	Priority weights for loads.
$\bar{\gamma}$	Most expected value of uncertain parameter in the IGDT approach.
R_c	System requirement in the IGDT approach.
S_b	Base load shedding in the IGDT approach.
β	Conservativeness degree in the IGDT approach.
M	A sufficiently large number in the DistFlow model.

Variables

x_{ij}	Line (i, j) connection status (1 = connected, 0 = otherwise).
u_i	DG connection status to bus i (1 = connected, 0 = otherwise).
σ_{ij}	Fictitious connection status of line (i, j) (1 = connected, 0 = otherwise).
λ_{ij}	Inclusion status of arc (i, j) in the directed spanning tree (1 = included, 0 = otherwise).
δ_{ij}	Parent-child status of line (i, j) (1 = if bus i is parent of bus j , 0 = otherwise).
f_{ij}^k	Flow of commodity k from bus i to bus j .
P_{ij}, Q_{ij}	Active and reactive power transfer between buses i and j .
P_s, Q_s	Upstream active and reactive power injection to slack bus.
P_i^G, Q_i^G	DG active and reactive output power.
P_i^S, Q_i^S	Active and reactive power curtailed at bus i .
V_i	Voltage magnitude at bus i .
X	Decision variables in the IGDT approach.
γ	Uncertain parameter in the IGDT approach.
α	Uncertainty radius in the IGDT approach.

Functions

$\hat{\alpha}(X, R_c)$	Robustness function in the IGDT approach.
$R(X, \gamma)$	System model in the IGDT approach.
$\Gamma(\bar{\gamma}, \alpha)$	Uncertainty set in the IGDT approach.

Algorithm Notations

n	Index of rows in LUT.
n_{max}	Maximum rows in LUT.
z	Index of number of DGs.
z_{max}	Maximum number of DGs.
C_H^{unit}	Unit cost of hardening distribution lines.
C_G^{unit}	Unit cost of allocating a backup DG in system.
B_P	Total planning budget.
$\hat{\beta}_z$	Best possible conservativeness degree for z DGs.
$\hat{\zeta}_z$	Minimum possible MWLS for z DGs.
$C_{P,z,n}$	Total cost of row n in LUT for z DGs.
$C_{H,z,n}$	Hardening cost of row n in LUT for z DGs.
$C_{G,z,n}$	DG allocation cost of row n in LUT for z DGs.

I. INTRODUCTION

ENHANCING the power system resilience against natural disasters has become a more essential effort due to the increasing dependency of critical infrastructures on electricity services. Historical data indicate that more than 30% of worldwide power outages from 1965 to 2012 were caused by natural disasters [1]. Hurricane Sandy in the U.S, widespread floods in Australia, and severe ice storms in China are such instances of disruptive events that have left about 7 million people, 150,000 customers, and 13 provinces without power, respectively [2]. Meanwhile, 90% of all power outages correspond to distribution networks [3], showing the significance of research at the distribution level.

Resilience in power systems is defined as the ability to predict severe events, adapt to or rapidly recover from these events, and acquire knowledge for long-term planning in operation to mitigate the effects of damages caused by the occurrence of such events in the future [2]. There are various approaches to retrieve power distribution systems from post-event situations. Distributed Generation (DG) resources are the most significant assets in distribution systems which can supply critical loads in emergency conditions; remotely controlled switches also play a key role in boosting the system robustness by isolating fault locations and reconfiguring the distribution network. However, given the high cost of distribution system hardening for extreme events and budget limitations, it is essential to consider optimal planning designs and operating conditions of distribution networks with respect to the inherent uncertainty of the resilience problem in natural disasters. Although resilience in power systems pertains to transmission and distribution levels, distribution systems, which are more vulnerable to disasters, have been more focused in recent years. Several resilience-oriented studies in distribution systems have exploited a deterministic model of extreme events in their methodologies. In [4], the uncertainty of intermittent energy resources was handled to restore critical loads in networked Microgrids (MGs) following

the loss of upstream substations. A Mixed-Integer Second-Order Cone Programming (MISOCP) model was proposed in [1] to minimize the load shedding by optimizing the DG islanding scheme as well as topology reconfiguration after a deterministic event resulting in multiple faults. In [5], the authors presented a decision-making method to coordinate multiple sources within MGs for load restoration with pre-supposed faulted lines. A Mixed-Integer Linear Programming (MILP) approach for MG formation was proposed in [6] that could pick-up high-priority loads in extreme events to improve the resilience of distribution systems. Authors in [7] developed a restoration approach to maximize served loads by means of dispatchable DGs, Renewable Energy Source (RES), and Energy Storage System (ESS), in a reconfigurable distribution grid. Although RES and load uncertainties were considered in the stochastic MILP model, the uncertainties of natural disasters were overlooked.

A resilient scheduling program was employed in [8] to reduce MG load curtailments in extreme events. In addition, uncertain consequences of extreme events on MG operations were modeled using a stochastic method, and merits of deploying MG facilities were exhibited. In [9], a two-stage stochastic Mixed-Integer Non-linear Programming (MINLP) model was proposed to optimize wind turbine allocations and topology reconfigurations, simultaneously, in distribution systems. Line outage scenarios following weather-related events were selected using a probabilistic scenario-based approach. However, this optimization problem does not yield an optimal hardening plan for distribution grids. In [10], the authors proposed a two-stage stochastic problem to make a Resilience-Oriented Design (ROD) with respect to catastrophic event uncertainties. Distribution line hardening, optimal DG placement, and automatic switch allocation were considered in the ROD by Distribution System Planners (DSPs). Furthermore, operation measures such as reconfiguration, load shedding, and DG resource re-dispatch were carried out at the post-fault stage.

A robust tri-level hardening plan was presented in [11] to improve the resilience of Integrated Electricity and Natural Gas Distribution Systems (IENDS). The model yields an optimal hardening plan for minimizing the total load shedding in IENDS. Authors in [12] proposed a tri-level mixed-integer min-max-min optimization model based on Defender-Attacker-Defender (DAD) game theory. The model was considered both planning and operation measures and used a linearized DistFlow model for power flow in distribution networks to mitigate malicious impacts of natural disasters. At the first level, optimal hardening was carried out considering the initial attack scenario; at the second level, the worst-case attack scenario was defined given the hardening plan; at the third level, operation measures were factored in to restore the distribution system. The objectives at the first and third levels were to reduce the weighted value of load shedding. In addition, critical loads were supplied in post-fault situations using multiple DGs, placed in the network. The optimal placement of DG resources is a vital element to make the distribution system more resilient, as shown in this paper. In [13], a two-stage Robust Optimization (RO) method

TABLE I
TAXONOMY OF RECENT RESILIENCE STUDIES IN DISTRIBUTION NETWORKS

Reference Number	Optimization Model	Power Flow	Topological Uncertainty				Configuration		DG Placement		Optimal Hardening Plan
			Not considered	Stochastic	RO	IGDT	Initial	Optimal	Initial	Optimal	
[1]	MISOCP	Relaxed PF	✓					✓	✓		
[6]	MILP	DistFlow	✓					✓	✓		
[7]	MILP	DistFlow	✓					✓	✓		
[8]	MILP	Piecewise linear PF		✓			✓		✓		
[9]	MINLP	AC-PF		✓				✓		✓	
[10]	MILP	DistFlow		✓				✓		✓	✓
[11]	MILP	DistFlow			✓		✓		✓		✓
[12]	MILP	DistFlow			✓			✓	✓		✓
[13]	MILP	DistFlow			✓		✓			✓	✓
This Paper	MILP	DistFlow				✓		✓		✓	✓

was employed to coordinate grid hardening and optimal DG allocation following a hurricane. However, network topology reconfiguration was not used to counter natural disasters. A summary of state-of-the-art resilience studies in distribution systems is provided in Table I, which includes optimization model, Power Flow (PF) model, and resilience measures.

To account for severe uncertainties, the application of Information Gap Decision Theory (IGDT) has recently been promoted in power systems. For instance, [14] and [15] adopted the IGDT to cope with load uncertainties. In [16], electricity prices and loads, as well as investment and operation costs, were modeled as uncertainty sources using the IGDT. A robust mixed-integer quadratically constrained programming model was proposed in [17] to facilitate distribution network restorations by applying the IGDT. The IGDT was also used to consider RES uncertainties. In [18], the IGDT was utilized to minimize the operating costs by solving a robust Unit Commitment (UC) problem. In this work, wind power generation uncertainty was taken into account to develop the IGDT performance functions, including risk-averse and risk-seeker strategies. Authors in [19] utilized the IGDT to simultaneously handle PV power and load uncertainties in an isolated MG in Africa. In the work, in addition to the IGDT, spinning reserve and ESSs were employed to cope with the RES variability.

As can be seen, most of the resilience-oriented studies have adopted probabilistic and RO techniques to deal with uncertainties in natural disasters. It has been shown in [20] that in addition to expensive computations and difficult implementation in complex models, these methods respectively need the availability of a large amount of historical data and known intervals to model topological uncertain parameters, such as distribution line outages in distribution systems. However, severe natural disasters (e.g., earthquakes) may not follow specific probability distribution functions, and there might not be sufficient information available to account for their inherent uncertainties. In contrast, the IGDT is an effective tool that can quantify severe uncertainties using an unknown uncertainty set, which does not require any specific information (i.e., probability density functions, membership functions, known intervals, etc.) [21]. Specifying a conservativeness degree is required as an essential input to implement an IGDT

approach [22], yet it is reflected from the system requirements and is independent of the uncertainty vector [20].

Although the IGDT has been applied to uncertain microeconomic and operational parameters [20], investigating its application in handling topological uncertainty—the one originated from extreme events—has remained untouched. In addition, a way by which the best conservativeness degree can be chosen has not been suggested in the literature to date. For instance, the next sections of this paper will demonstrate that the best conservativeness degree can be selected based on an available planning budget and trade-off analysis.

Consequently, this paper sheds light on a novel IGDT-based resilience planning model for distribution networks in the face of natural disasters, providing the following contributions:

- Application of the IGDT method is expanded to topological uncertainties to develop a potent resilience tool.
- A robustness function is presented to determine risk-averse regions of distribution line outages as well as their total damaged lengths.
- An optimal hardening plan, optimal DG allocation, and optimal topology reconfiguration are proposed through a linear optimization model to ensure the supply of a pre-defined level of loads following a natural disaster with devastating impacts.
- A novel conservativeness degree selection algorithm is proposed to tune the model inputs while simultaneously determining an optimal trade-off between the number of DGs and the length of lines hardened.

The rest of this paper is organized as follows: The proposed IGDT-based network planning model is presented in Section II. Section III describes the steps required to implement the model, and provides the proposed conservativeness degree selection algorithm. Case studies and simulation results are provided in Section IV, and finally, Section V concludes the paper.

II. PROPOSED IGDT-BASED RESILIENCE MODEL

Despite reliability definitions, resilience is characterized to assess High Impact Low Probability (HILP) events [23]. Probabilistic methods may be unable to provide effective results for events that are less probable, even if the requirement of large historical data for implementing such methods was not

strenuous. This would highlight the necessity of implementing robust strategies against extreme events. While both the IGDT and RO methods can offer robust decisions [14], they are fundamentally different in modeling. In order to utilize an RO method in resilience studies, one salient point is to assume an intensity level for extreme events, referred to as known intervals. For instance, the number of line outages caused by an extreme event is an essential input for an RO method, known as disaster severity level, attack budget, etc [11]–[13]. On the other hand, the robustness feature of the IGDT approach is to maximize that disaster severity level while immunizing the system against an unfavorable increase in the load shedding value. In an IGDT approach, not only the maximum damage scenario is obtained, but also it is guaranteed that the load shedding level would not exceed a desired pre-defined threshold, specified by a conservativeness degree.

Generally speaking, uncertainty can be regarded as either beneficial or adverse. While the propitiousness of uncertainty entails an unimagined success, its perniciousness will increase the possibility of failure. One particularly salient example of this is weather condition as an uncertain parameter. While a sunny day with an unexpected higher level of solar irradiation is rewarding for photovoltaic generation, a stormy day can be damaging for electrical infrastructures, such as distribution systems. These potentials are dealt with immunity functions in IGDT principles, which are called opportuneness and robustness functions. In an IGDT approach, a risk-averse strategy is adopted through a robustness function to address the consequences of the negative side of the uncertainty. Natural disasters are inherently destructive events that can significantly impose high levels of risk on electrical networks. A risk-seeker strategy through an opportuneness function, on the other hand, intends to attain more profit from the uncertain parameters and, hence, is not applicable to reducing the risks arisen from those HILP events. In this paper, we will show how by manipulating the risk-averse strategy, a new framework can be developed through a robustness function to overcome the negative impacts of extreme events. Note that the word ‘robust’ used in the rest of this paper stems from the IGDT robustness function and is not related to the RO method.

In the following, we first briefly review the pure formulation of the risk-averse IGDT and, subsequently, explain how the uncertainty of natural disasters can be expressed through an IGDT approach.

A. The IGDT: General Formulation

In this section, the general representation of uncertainty in the risk-averse strategy is provided and discussed. In this regard, the uncertainty model is formulated as follows [21]:

$$\Gamma(\bar{\gamma}, \alpha) = \left\{ \gamma : \left| \frac{\gamma - \bar{\gamma}}{\bar{\gamma}} \right| \leq \alpha \right\} \quad (1)$$

where Γ is an uncertainty set that describes the behavior of uncertain parameters; γ indicates the uncertain parameter and $\bar{\gamma}$ is its most expected value. Hence, α presents the uncertainty radius (i.e., maximum possible deviation of γ). Let us first

define the following problem:

$$\hat{\alpha}(X, R_c) \triangleq \max\{\alpha\} \quad (2)$$

Subject to:

$$\max R(X, \gamma) \leq R_c \quad (3a)$$

$$\gamma \in \Gamma(\bar{\gamma}, \alpha) \quad (3b)$$

$$X \in \mathcal{D} \quad (3c)$$

where X is the set of decision variables and $\mathcal{D}$ is its domain; $R(X, \gamma)$ signifies the system model, and R_c is a general representation that shows the satisfaction of the minimum system requirements through (3a). This constraint will be explained in more details next. Equation (2) states that the robustness function is the maximum deviation level of the uncertain parameter from its most expected value. Constraint (3a) ensures that the maximization procedure due to the robustness function will still satisfy all the system constraints. However, this model is a very generic representation of the risk-averse strategy through the robustness function. Now let us rephrase the problem in the following:

$$\max_{X \cup \alpha} \alpha \quad (4)$$

Subject to:

$$g_p(X, \gamma) \leq 0, p = \{1, 2, \dots, p\} \quad (5a)$$

$$h_q(X, \gamma) = 0, q = \{1, 2, \dots, q\} \quad (5b)$$

$$f(X, \gamma) \leq (1 + \beta)f_0 \quad (5c)$$

$$\gamma \in \Gamma(\bar{\gamma}, \alpha) \quad (5d)$$

$$X \in \mathcal{D} \quad (5e)$$

where $g_p(X, \gamma)$ and $h_q(X, \gamma)$ denote the system inequality and equality constraints, respectively, and p and q signify their numbers. Indeed, (3a) has been decomposed into (5a) to (5c), which include the system constraints plus one additional constraint to be satisfied, as suggested by [22]. β is a conservativeness degree that controls the deviation of some system variables due to the maximization of the uncertainty radius; in this work, f is a specific system variable whose value should remain below the β -controlled threshold, and f_0 denotes its basic value. The application of these constraints will be explained in the following section.

B. The IGDT: Resilience Application

Extreme events (e.g., earthquakes, ice storms, etc.) are uncertain phenomena that are hard to predict. Even the availability of a large amount of historical data may not properly predict their occurrence time and intensity. It has been shown that almost all the damaging impacts of disruptive events are in distribution networks [3], and distribution lines are direct victims of such events. We do not know how many distribution lines will be damaged in a likely natural disaster because its intensity and occurrence time is vague and uncertain. It has been suggested in [20] that distribution line outages can be an informative indicator for quantifying the uncertainty of extreme events. Now let us define the following.

γ : the length of distribution line outages that will be originated from an extreme event.

$f(X, \gamma)$: the value of load shedding that will be caused by those line outages.

It is evident that $f(X, \gamma)$ should be a function of γ and other system variables X ; γ can, therefore, be mathematically expressed as follows:

$$\gamma = \frac{1}{2} \sum_{i \in \mathbf{B}} \sum_{j \in \mathbf{B}} C_{ij} \ell_{i,j} - \frac{1}{2} \sum_{i \in \mathbf{B}} \sum_{j \in \mathbf{B}} x_{ij} \ell_{i,j} \quad (6)$$

where x_{ij} denotes a binary variable indicating whether the line connecting nodes i and j is in-service (1 if connected); $C_{i,j}$ presents the set of candidate lines in a distribution network, including normally closed lines and tie lines, and $\ell_{i,j}$ is the length of each line.

As previously mentioned, $\bar{\gamma}$ is the most expected value of γ . Therefore, it is defined as the maximum outage length by which a portion of loads can be supplied by the *entire* capacity of energy sources (e.g., DGs) located in the network. For example, assume an islanded distribution network with a total demand of 1,000 kW. It is planned to place a 200 kW DG resource in the network to supply a portion of loads. If the entire 200 kW capacity of the DG is used to supply 200 kW of neighboring loads, the maximum possible outage length is assumed to be 5 km. If only 40% of the DG's capacity is utilized to supply 80 kW of neighboring loads, the maximum possible outage length can reach 8 km, but obviously the load shedding level escalates, as well. In this example, therefore, $\bar{\gamma} = 5$ km and $\gamma = 8$ km.

From the mathematical point of view, the maximizing operator in (4) enlarges the radius of uncertainty presented in (1), so that γ reaches its maximum value while (5a) to (5e) are satisfied.

From the resilience viewpoint, on the other hand, more intense natural disasters are expected to result in more line outages, contributing similarly to a bigger γ . Therefore, this model can be interpreted as follows: There are a number of operating equality/inequality constraints, such as power flow and radiality constraints, that should be satisfied in (5a) and (5b); we then try to maximize the length of lines on outage via (4) while keeping the load shedding level below a threshold specified by the conservativeness degree β in (5c). When the maximum out-of-service lines are determined, the remaining lines are responsible to flow the power to priority loads. Admittedly, the survival of these lines is an essential attempt that will be discussed in the following section.

DSPs typically wish to minimize the load shedding level following a natural disaster. There may be several emergency, critical loads in every distribution network that should receive higher priorities to be supplied. Implementing a model to evaluate the maximum allowable outage length in a distribution system, subject to supplying the critical loads, enables us to prepare the network for the worst-case attack scenario taken place in the future. In other words, we can restrict the model to supply some system loads considering their priority weights, with the minimum in-service lines to provide a robust framework through the IGDT robustness function. Constraint (5c)

can, therefore, be rewritten as follows:

$$\sum_{i \in \mathbf{B}} \omega_i P_i^S \leq (1 + \beta) S_b \quad (7)$$

where S_b denotes the basic value of load shedding, and $\omega_i P_i^S$ is the priority weighted load shedding value at node i . In the above-mentioned example, $S_b = 1,000 - 200 = 800$ kW.

C. Resilience Measures

In this section, it is shown that how we gain maximum benefit from the aforementioned model to develop a potent resilience tool. In the following, two effective resilience measures are presented:

1) *Line Hardening*: Through the previous section, we showed that how we are able to maximize the damage scenario while meeting a minimum system load level. Since the total length of available lines (i.e., candidate lines) in a distribution network is constant, by maximizing the length of damaged lines, the length of remaining, in-service lines is inevitably minimized. If we can make these minimum in-service lines resilient against extreme events, then we can ensure that the most of high-priority loads are incessantly supplied through these sustainable paths. Hardening these lines with high quality materials is a resilience measure that improves the robustness feature of the distribution system against extreme events [10]. In numerous studies, hardening has been carried out as an effective strategy in distribution systems [11]–[13], [24], [25]. Some research works suggest the vegetation management as the primary operational technique to enhance the network resilience [26]. Breaking and falling trees by weather-related events can seriously threaten near-ground distribution conductors. Accordingly, vegetation management and trimming can reduce the risk of falling trees into utility lines. Augmenting the structural performance of utility systems is the other effective hardening technique. In practice, upgrading poles to higher classes can help better resist against mechanical stresses during events [27]. This study entails both the vegetation management and pole upgrade techniques in the hardening strategy and, consequently, incorporates their individual implementation costs in the total planning budget, described in Section III. Thus, it is assumed that hardened lines remain intact after natural disasters [11]–[13].

2) *DG Allocation*: Hardening lines without the availability of a reliable power supply in the aftermath of an extreme event can be a squander of money since upstream substations may not provide sufficiently reliable services. There are numerous instances showing that during a natural disaster the upstream substation is disrupted due to, for example, the operation of protection systems or damaged transmission infrastructures [4]–[6]. The system resilience can, therefore, be enhanced by the application of DGs. DG resources, which have increasingly penetrated into distribution networks in recent years [1], [6]–[13], [28], are known as rapid responding units for supplying high-priority loads in emergency conditions [11]. The on-site DG units with diverse input resources, e.g., fossil fuels, account for one of the efficient smart grid technologies for enhancing the resilience [1]. In this paper, it is assumed

that backup DG resources are committed to picking up some system loads in the aftermath of a natural disaster. In the normal operation of distribution networks, the local demand is generally met through upstream substations due to their usually lower operational costs. Along with the upstream grid, RESs may be employed to account for a portion of the power supply. RESs, however, are intermittent power units suffering typically from severe uncertainties [29]. On the other hand, in a resilience-oriented planning, highly reliable power sources should be utilized to curb load shedding risks in post-fault situations. Therefore, fossil fuel DGs could be a more desirable choice in the context of resilience. Recently, fossil fuel DGs have been manipulated in the form of mobile generators [30], as well. In addition to providing ancillary services in normal operating states of the system, they can also be pre-placed in their optimal locations prior to a forecasted natural disaster to mitigate outage risks. In this paper, the proposed approach, however, is tailored for totally unexpected extreme events, where there exists insufficient information regarding their occurrence time and intensity.

D. System Constraints

In this section, inequality, equality, and domain constraints that describe (5a), (5b), and (5e) are presented.

1) *Power Flow*: In an electric distribution network, DG resources and other network components should be considered in power flow equations. Several papers have adopted the linear approximation of distribution network power flow in optimization problems [10]–[13]. The linear DistFlow model has also been used in MG studies [6], [7]. Therefore, in this paper, the linearized DistFlow equations are used, as follows:

$$V_j + \frac{R_{ij}P_{ij} + X_{ij}Q_{ij}}{V_0} - (1 - x_{ij})M - V_i \leq 0 \quad \forall (i, j) \in \mathbf{L} \quad (8)$$

$$V_i - \frac{R_{ij}P_{ij} + X_{ij}Q_{ij}}{V_0} - (1 - x_{ij})M - V_j \leq 0 \quad \forall (i, j) \in \mathbf{L} \quad (9)$$

where the big M method is adopted to guarantee that voltages at two disconnected buses are separated [7], [10].

$$\sum_{j \in \mathbf{B}} P_{ij} - P_s - P_i^S - P_i^G + P_i^D = 0 \quad \forall i \in \mathbf{B} \quad (10)$$

$$\sum_{j \in \mathbf{B}} Q_{ij} - Q_s - Q_i^S - Q_i^G + Q_i^D = 0 \quad \forall i \in \mathbf{B} \quad (11)$$

where (10) and (11) present active and reactive power balance constraints.

2) *Load Shedding Constraints*: The following constraints outline the limits associated with both active and reactive load curtailments:

$$P_i^S - P_i^D \leq 0 \quad \forall i \in \mathbf{B} \quad (12)$$

$$Q_i^S - Q_i^D \leq 0 \quad \forall i \in \mathbf{B} \quad (13)$$

$$P_i^S Q_i^D - P_i^D Q_i^S = 0 \quad \forall i \in \mathbf{B} \quad (14)$$

$$P_i^S \in \mathbb{R}_+ \quad \forall i \in \mathbf{B} \quad (15)$$

$$Q_i^S \in \mathbb{R}_+ \quad \forall i \in \mathbf{B} \quad (16)$$

where (12) and (13) declare that active and reactive load shedding at bus i cannot exceed the corresponding load value.

Furthermore, active and reactive loads are curtailed at the same ratio as shown in (14).

3) *Voltage and Line Flow Limits*: Lower and upper limits of the bus voltage magnitude are specified in (17) and (18), respectively.

$$V_i^{\min} - V_i \leq 0 \quad \forall i \in \mathbf{B} \quad (17)$$

$$V_i - V_i^{\max} \leq 0 \quad \forall i \in \mathbf{B} \quad (18)$$

Furthermore, active and reactive power flow constraints of distribution lines are demonstrated as follows:

$$P_{ij} - S_{ij}^{\max} x_{ij} \leq 0 \quad \forall (i, j) \in \mathbf{L} \quad (19)$$

$$-S_{ij}^{\max} x_{ij} - P_{ij} \leq 0 \quad \forall (i, j) \in \mathbf{L} \quad (20)$$

$$Q_{ij} - S_{ij}^{\max} x_{ij} \leq 0 \quad \forall (i, j) \in \mathbf{L} \quad (21)$$

$$-S_{ij}^{\max} x_{ij} - Q_{ij} \leq 0 \quad \forall (i, j) \in \mathbf{L} \quad (22)$$

$$P_{ij} \in \mathbb{R} \quad \forall (i, j) \in \mathbf{L} \quad (23)$$

$$Q_{ij} \in \mathbb{R} \quad \forall (i, j) \in \mathbf{L} \quad (24)$$

4) *Backup DGs*: The power generation limits of backup DGs are formulated as follows:

$$P_i^G - P_{\max}^G u_i \leq 0 \quad \forall i \in \mathbf{B} \quad (25)$$

$$Q_i^G - Q_{\max}^G u_i \leq 0 \quad \forall i \in \mathbf{B} \quad (26)$$

$$Q_{\min}^G u_i - Q_i^G \leq 0 \quad \forall i \in \mathbf{B} \quad (27)$$

$$P_i^G \in \mathbb{R}_+ \quad \forall i \in \mathbf{B} \quad (28)$$

$$Q_i^G \in \mathbb{R} \quad \forall i \in \mathbf{B} \quad (29)$$

where u_i is a binary variable to show whether a backup DG resource is connected to bus i (1 if connected).

5) *Radial Operation*: A distribution network with a mesh topology could be radially operated [12]. Following a natural disaster, causing major power line outages, a reconfiguration is carried out for a fast service restoration. Meanwhile, the radial configuration is often maintained to facilitate the constrained operation and protection device settings [31]. In practice, there exist various approaches in the literature toward the imposition of radiality constraints in distribution networks. Spanning tree and single-commodity flow models are two prevailing radiality methods presenting mathematical formulations applicable to MILP problems. With the flexibility of the number of islands as well as the number of DGs within MGs, the former guarantees that each distribution node has no more than one parent [7]. However, counterexamples imply that this parent-child model might produce subgraphs with loops [32], [33]. The latter, on the contrary, ensures the radial topology of islands in the post-fault period, yet the number of MGs is fixed and the model is limited to allocating more than one DG in each island [34]. Recently, a novel approach has been developed to formulate radiality constraints for resilient formation of MGs [35]. The proposed directed multicommodity flow-based model enables several flexibilities of distribution systems, including the number of MGs and the allocation of DGs into MGs. Besides, it allows the intentional formation of unenergized islands, enabling the optimization procedure to flexibly pick-up loads. Therefore, in this paper, we employ formulations from [35] to guarantee the radial topology of flexible

islands while ensuring the radial operation of potential connected DGs in each separate MG by adopting the parent-child constraints [7]. The radiality constraints are given as follows:

$$\sum_{(j,i_s) \in \mathbf{L}} f_{ji_s}^k - \sum_{(i_s,j) \in \mathbf{L}} f_{i_sj}^k + 1 = 0 \quad \forall k \in \mathbf{B} \setminus i_s \quad (30)$$

$$\sum_{(j,i) \in \mathbf{L}} f_{ji}^k - \sum_{(i,j) \in \mathbf{L}} f_{ij}^k = 0 \quad \forall k \in \mathbf{B} \setminus i_s, \forall i \in \mathbf{B} \setminus \{i_s, k\} \quad (31)$$

$$\sum_{(j,k) \in \mathbf{L}} f_{jk}^k - \sum_{(k,j) \in \mathbf{L}} f_{kj}^k - 1 = 0 \quad \forall k \in \mathbf{B} \setminus i_s \quad (32)$$

$$f_{ij}^k - \lambda_{ij} \leq 0 \quad \forall k \in \mathbf{B} \setminus i_s, \forall (i,j) \in \mathbf{L} \quad (33)$$

$$f_{ji}^k - \lambda_{ji} \leq 0 \quad \forall k \in \mathbf{B} \setminus i_s, \forall (j,i) \in \mathbf{L} \quad (34)$$

$$f_{ij}^k, f_{ji}^k \in \mathbb{R}_+ \quad \forall k \in \mathbf{B} \setminus i_s, \forall (i,j) \in \mathbf{L} \quad (35)$$

$$\sum_{(i,j) \in \mathbf{L}} (\lambda_{ij} + \lambda_{ji}) - |\mathbf{B}| + 1 = 0 \quad (36)$$

$$\lambda_{ij} + \lambda_{ji} - \sigma_{ij} = 0 \quad \forall (i,j) \in \mathbf{L} \quad (37)$$

$$x_{ij} - \sigma_{ij} \leq 0 \quad \forall (i,j) \in \mathbf{L} \quad (38)$$

$$\delta_{ij} + \delta_{ji} - x_{ij} = 0 \quad \forall (i,j) \in \mathbf{L} \quad (39)$$

$$\sum_{i \in \mathbf{B}} \delta_{ij} - 1 \leq 0 \quad \forall j \in \mathbf{B} \quad (40)$$

$$\delta_{ij} \in [0, 1] \quad \forall (i,j) \in \mathbf{L} \quad (41)$$

where (30)-(38) represent the flow-based topological radiality constraints, and (39)-(41) enable the radial operation of multiple DGs. To form a radial topology, in the first step, (30)-(37) are employed to form a fictitious spanning tree. Secondly, constraint (38) is utilized to provide a flexible reconfiguration by choosing a subgraph of the fictitious spanning tree in the previous step. Detailed explanations of these constraints are provided in [35]. In a radially operated MG, each connected line has one parent and one child node. This parent-child status is determined by δ_{ij} and δ_{ji} . Equation (39) implies that if the line between nodes i and j is connected, either i is the parent of j ($\delta_{ij} = 1$) or j is the parent of i ($\delta_{ji} = 1$). Moreover, according to (40), each node in a MG has no more than one parent node.

6) *Topology Reconfiguration*: In addition to the above mentioned practices, the DSP can take further soft measures to resist unexpected outages [36]. In pre-fault conditions, the tie lines are open and the system demand is met through the upstream substation. However, a natural disaster may cause multiple faults and disrupt the operation of the upstream network. In such a situation, the network may be divided into several islands; in some islands, which have a backup DG resource, even though the main network is lost, the demand may be picked up by local DGs. The load shedding will occur in islands, which have neither DGs nor tie lines. Other islands can be powered by reconfiguring their ties with DG-powered areas following a fault [1]. In this study, the most desirable configuration, through which the critical loads are supplied, are found according to the following topology reconfiguration constraints:

$$\sigma_{ij} - C_{ij} \leq 0 \quad \forall (i,j) \in \mathbf{L} \quad (42)$$

$$x_{ij} - x_{ji} = 0 \quad \forall (i,j) \in \mathbf{L} \quad (43)$$

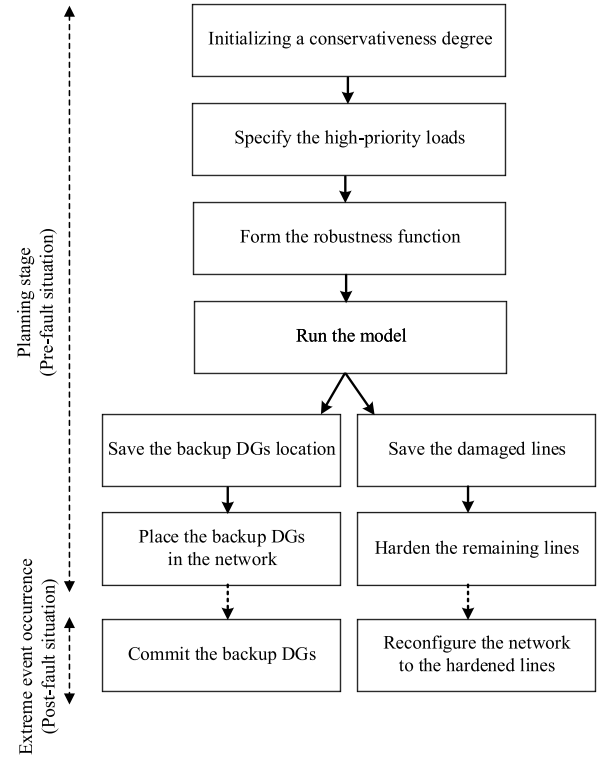


Fig. 1. Proposed IGDT-based resilience plan.

where constraint (42) declares that the previously discussed fictitious radial topology and, eventually, the post-fault configuration of distribution lines (i.e., the set of in-service lines) are determined based upon the mesh topology of the system under study, which includes normally closed branches as well as tie lines. Additionally, (42) incorporates the possibility of hardening tie lines in the model. Equation (43) is also self-explanatory, designating x_{ij} as an undirected variable.

III. MODEL IMPLEMENTATION

A. Steps

In the previous section, we formulated the proposed IGDT-based resilience planning model. The steps required to implement the resilience scheme is depicted in Fig. 1. At first, a suitable value of β is chosen by the DSP. The procedure to find the most suitable β is provided next. At this point, the priority loads are selected, and then, the maximum damage scenario is obtained through the IGDT robustness function. The robustness function yields the optimal location of backup DGs in addition to the line outage status in the network. In the planning stage, the DSP places the backup DGs in their optimal locations and hardens those lines that should be in-service in post-fault conditions. In an extreme event, the operator will commit the backup DGs and reconfigure the network topology based on the hardened lines. While the robustness function guarantees that the load shedding level will not exceed a pre-defined value, the actual load shedding level could be even lower if some unhardened lines survive in extreme events, as well.

Algorithm 1 Conservativeness Degree Selection

```

1: Initialization:  $z \leftarrow 1$ ;
2: while  $z \leq z_{max}$  do
3:    $n \leftarrow 1$ ;
4:   Calculate  $C_{G_{z,n}}$ ;
5:   Solve the MILP problem with the given number of
     DGs;
6:   Generate the LUT with 100 different  $\beta$  values;
7:   while  $n \leq n_{max}$  do
8:     Select plan number  $n$  from the LUT;
9:     Calculate  $C_{H_{z,n}}$ ;
10:    if  $C_{H_{z,n}} \leq B_P - C_{G_{z,n}}$  then
11:       $\hat{\beta}_z = \beta$ ;  $\hat{\zeta}_z = \text{MWLS} = (1 + \beta)S_b$ ;
12:      Save the planning scheme;
13:      break;
14:    end if
15:  end while
16: end while
17: Find the minimum  $\hat{\zeta}_z$ ;
18: return the corresponding  $z$ ,  $\hat{\beta}_z$ , and the planning scheme;

```

B. Conservativeness Degree Selection

In this section, an iterative algorithm is presented to facilitate the decision-making procedure in the proposed model. As previously stated, the IGDT method needs to receive a specific conservativeness degree as an input to solve the MILP problem. According to (7), it can be observed that the lower the chosen β , the lower the value of Maximum Weighted Load Shedding (MWLS) in post events. In practice, the lowest MWLS would occur when β is zero by utilizing the entire capacity of DGs in MGs. However, since DSPs are provided with a limited planning budget, it is of critical importance to successfully cope with resilience-oriented strategies in terms of quantity. DG resource allocation and distribution line hardening are the two planning strategies encompassed in the total budget. Given the monetary budget for the resilient planning of a distribution system, the DSP should decide whether to allocate more backup DGs or invest more budget in the line hardening strategy. The proposed decision-making procedure for optimal budget allocation as well as the best possible conservativeness degree selection is provided in Algorithm 1. First, with a given number of DGs (i.e., z), the proposed MILP problem is solved for 100 different values of conservativeness degree—a large enough number for covering as much scenarios as possible; then, a Lookup Table (LUT) is generated accordingly. A shortened instance of an LUT is given in Table III, which shows β , MWLS, total outage length, and the optimal hardening scheme, for different planning decisions. Selecting the rows of LUT through a top-down approach by commencing with $n = 1$, if the budget limitation is satisfied in plan number n , its corresponding information in LUT is saved. Note that the lower the n , the lower the MWLS, the higher the required hardening budget. At this point, another DG is added to the network and the process starts anew. Note that the above process is carried out for different number of DGs, ranging from $z = 1$ (i.e., one DG) to the maximum

TABLE II
UNIT COST OF IMPLEMENTING RESILIENCE-ORIENTED STRATEGIES

No.	Strategy	Unit	Value
1	Line hardening	\$/km	141,322 [25]
2	Backup DG allocation	\$/kW	1,000 [37]

number of DGs, z_{max} , which is determined by (47). Finally, comparing the saved plans for each z , the plan with the lowest MWLS will define the proper β to be chosen with z DGs in the network.

$$C_{P_{z,n}} = C_{H_{z,n}} + C_{G_{z,n}} \quad \forall z, n \quad (44)$$

$$C_{H_{z,n}} = \frac{1}{2} \sum_{(i,j) \in \mathbf{L}} (x_{ij,z,n} \ell_{i,j}) C_H^{unit} \quad \forall z, n \quad (45)$$

$$C_{G_{z,n}} = z C_G^{unit} \quad \forall z, n \quad (46)$$

$$z_{max} = \left\lceil \frac{B_P}{C_G^{unit}} \right\rceil \quad (47)$$

where $C_{P_{z,n}}$, $C_{H_{z,n}}$, and $C_{G_{z,n}}$ denote the total planning, hardening, and DG allocation costs for plan number n in the LUT with z DGs, respectively; B_P is the total monetary budget, and C_G^{unit} and C_H^{unit} respectively show the cost of installing one backup DG in the system and the cost of line hardening per one kilometer.

IV. NUMERICAL EXAMPLES**A. Solution Procedure**

The proposed resilience model consists of linear constraints stated in the DistFlow model and integer variables presented in a MILP problem. The model was coded in GAMS environment [38] and solved by the commercial solver CPLEX 12.6. The results were obtained using an Intel core i5-3210M CPU 2.50 GHz PC with 6 GB of RAM in approximately 12 minutes for the 33-bus test system and 52 minutes for the 94-bus test system, which are total execution times for deriving the IGDT robustness function of 100 different values for β . Admittedly, the computation time is not generally a significant issue in planning problems since the DSPs are provided with enough time to make appropriate decisions in the long run. In order to assess the quality of solutions, it is worth mentioning that an additional option `AIJOptcr = 0` was used in the GAMS environment to ensure that the results are all global optimal [39].

B. IEEE 33-Bus Distribution Network

1) *Test System*: In this section, computational experiments are performed on an IEEE 33-node distribution network to illustrate the performance of the proposed model. The distribution system has five tie lines and one upstream substation. It is assumed that the available monetary budget is \$2,500,000, and the total planning cost is calculated according to Table II, showing the cost of implementing each resilience-oriented strategy. It is also presumed that controllable backup DGs are identical natural gas-fired CHPs, each with a capacity of 400 kW [37]. The upper and lower levels of voltage magnitude

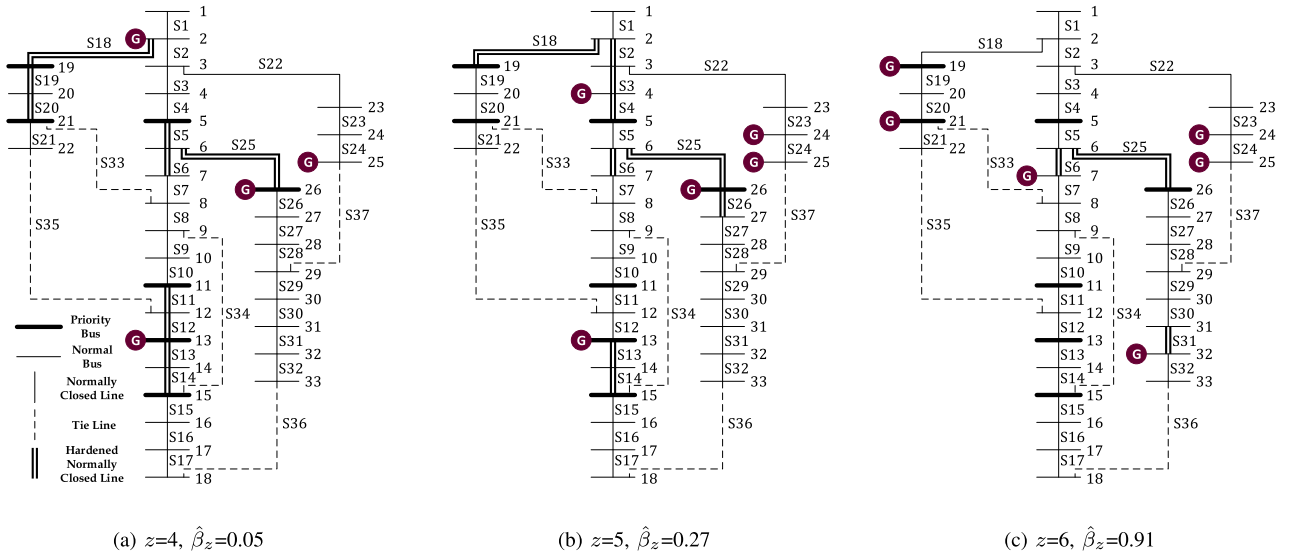


Fig. 2. Optimal planning schemes for different number of DGs in the 33-bus distribution test network.

TABLE III
A SHORTENED LUT WITH FIVE DGs IN THE 33-BUS
DISTRIBUTION TEST NETWORK

n	β	MWLS (kW)	Length of outages (km)	Optimal hardening scheme	$C_{H5,n}$ (\$)
1	0.02	1,749.30	20.35	5, 6, 10, 11, 12, 13, 14, 18, 19, 20, 25, 26	953,923.5
4	0.07	1,835.05	20.85	5, 6, 11, 12, 13, 14, 18, 19, 20, 25	883,262.5
8	0.14	1,955.10	21.95	5, 6, 13, 14, 15, 18, 19, 20, 25	727,808.3
12	0.20	2,058.00	22.70	5, 6, 13, 14, 18, 19, 20, 25	621,816.8
16	0.27	2,178.05	23.90	2, 3, 4, 6, 13, 14, 18, 25, 26	452,230.4
22	0.38	2,366.70	25.05	2, 3, 4, 6, 18, 25, 31	289,710.1
30	0.51	2,589.65	25.75	2, 6, 18, 25, 31	190,784.7
40	0.68	2,881.20	26.70	6, 25	56,528.8
48	0.82	3,095.00	27.10	-	0

TABLE IV
THE MINIMUM POSSIBLE MWLSs FOR DIFFERENT NUMBER OF DGs
IN THE 33-BUS DISTRIBUTION TEST NETWORK

z	$\hat{\zeta}_z$ (kW)	$\hat{\beta}_z$	$C_{Pz,n}$ (\$)	B_P (\$)
2	2,915.00	0	1,852,848.9	2,500,000
3	2,515.00	0	2,224,584.5	2,500,000
4	2,220.75	0.05	2,483,262.5	2,500,000
5	2,178.05 *	0.27	2,452,230.4	2,500,000
6	2,511.65	0.91	2,498,925.4	2,500,000

* Optimal value of MWLS.

are set to 1.1 and 0.9 p.u., respectively. The line flow capacity is also 5 MVA. The rest of the system data are available at [40]. Moreover, the data for the length of distribution lines are available in [41].

2) *Base Case*: As aforementioned, it is assumed that the upstream transformer is disconnected following an extreme event, and a portion of loads in the distribution network is supplied by the backup DGs. In the base case, β is zero and the uncertain parameter γ equals its most expected value, denoted by $\bar{\gamma}$. Therefore, the base load shedding is obtained based on the number of DGs located in the network. The total demand of the 33-bus test system is 3,715 kW. For instance, if five 400 kW DGs are used to supply the loads, the base load shedding S_b would be $3,715 \text{ kW} - 5 \times 400 \text{ kW} = 1,715 \text{ kW}$, and its corresponding $\bar{\gamma}$ is 19.85 km. Calculating the optimal number of DGs is discussed next.

3) *Conservativeness Degree Selection*: This section intends to determine the best possible β as well as the optimal number of DGs for the 33-bus distribution network. For this aim, the

model is solved for different number of DGs, ranging from $z = 1$ to $z = 6$, each with 100 different values of β ; note that there is no restriction on the maximum level of the conservativeness degree because larger values of β are automatically not selected by the algorithm. When the model is solved for each z , the results are saved in an LUT with 100 rows. For example, for five DGs in the network, a shortened LUT is shown in Table III. It is observed that a lower n results in a lower MWLS but a higher hardening cost. Since the total planning budget is limited, the lowest n may not be selectable, depending on the number of DGs purchased. For example, the total planning budget is considered \$2,500,000, and the cost of five DGs is $5 \times 400 \text{ kW} \times 1,000 \text{ $/kW} = \$2,000,000$. The maximum possible cost of hardening is, therefore, \$500,000. Now again by taking a look at Table III, it is observed that the first possible row in the table comes with $n = 16$, which has a hardening cost of \$452,230.4. At this point, the best possible conservativeness degree for five DGs is $\hat{\beta}_5 = 0.27$ and the minimum possible MWLS is $\hat{\zeta}_5 = 2178.05 \text{ kW}$. This process is carried out for all number of DGs, and the results are accordingly provided in Table IV. Note that the results for $z = 1$ is not shown in the table because only one 400 kW DG is unable to meet the critical loads. Among the minimum possible MWLSs of each number of DG, the lowest one determines the optimal number of DGs. In Table IV, therefore, 2,178.05 kW is the lowest one, so $z = 5$ is the optimal number of

TABLE V
A SHORTENED LUT FOR SIX DGs IN THE 94-BUS DISTRIBUTION TEST NETWORK

n	β	MWLS (kW)	Length of outages (km)	Optimal hardening scheme	$C_{H_0,n}$ (\$)
2	0.03	16,840.5	37.60	3-4, 4-5, 5-6, 5-55, 6-7, 7-9, 7-60, 12-13, 16-26, 16-17, 17-18, 18-19, 19-20, 20-83, 25-26, 26-27, 27-28, 28-32, 31-32, 32-33, 33-34, 34-46, 45-46, 50-51, 51-52, 52-53, 53-54, 54-55, 58-59, 59-60, 71-72, 78-79, 79-80, 80-81, 81-82, 82-83	2,649,787.5
5	0.09	17,821.5	43.00	3-4, 4-5, 5-6, 5-55, 6-7, 7-9, 7-60, 12-13, 16-17, 16-26, 25-26, 26-27, 27-28, 28-32, 32-33, 33-34, 34-35, 34-46, 45-46, 50-51, 51-52, 52-53, 53-54, 54-55, 58-59, 59-60, 60-61, 71-72	1,794,789.4
10	0.17	19,129.5	48.75	3-4, 4-5, 5-6, 6-7, 7-9, 7-60, 12-13, 31-32, 32-33, 33-34, 34-46, 45-46, 58-59, 59-60, 60-61, 71-72	1,074,047.2
20	0.34	21,909.0	54.80	6-7, 7-60, 12-13, 58-59, 59-60, 71-72	219,049.1
30	0.51	24,667.5	56.35	-	0

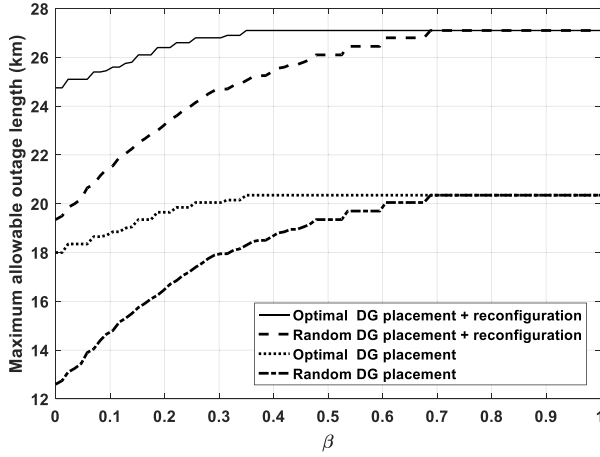


Fig. 3. Robustness function with five DGs for different cases.

DGs that can be placed in the network to have the minimum load shedding while meeting the budget limits, with β equal to 0.27. Now turning back to Table IV, the hardening scheme for $\beta = 0.27$ is utilized to strengthen the selected lines.

4) *Optimal Planning Scheme*: In this section, the optimal hardening scheme is presented for five DGs and $\beta = 0.27$. For the sake of comparison, the results for four and six DGs are provided considering their best conservativeness degrees. Fig. 2 depicts the hardening schemes for these three selected DG numbers. In this figure, DGs optimal locations, network topology, and priority loads are shown. Note that ω_i is 3 for priority loads, distinctly illustrated via a bold busbar and is 1 for the rest of the loads. For instance, while all the priority loads are within energized MGs for $z = 4$, loads at buses 11 and 21 are not picked up for $z = 5$. This is because even though backup DG resources can supply their adjacent buses through their adjacent hardened lines, by adding one backup DG, the DG allocation cost increases, and only a few lines can be hardened to satisfy the total planning budget. As a result, the model, for example, ignores to harden S11 and S12 with a hardening cost of \$261,445.7, for the satisfaction of only 45 kW high-priority demand at node 11. Additionally, even considering their three times larger weights, high-priority buses may be dominated by significant load levels. For instance, the 420 kW demand at bus 24 is supplied by the added DG in $z = 5$, which is much higher than the 45 or 90 kW loads located at the priority buses 11 and 21, respectively. Note that if these 45 or 90 kW loads must be supplied under every circumstances, one can consider a significantly

TABLE VI
THE MINIMUM POSSIBLE MWLSS FOR DIFFERENT NUMBER OF DGs IN THE 94-BUS DISTRIBUTION TEST NETWORK

z	$\hat{\zeta}_z$ (kW)	$\hat{\beta}_z$	$C_{P_z,n}$ (\$)	B_P (\$)
4	20,350.00	0	12,869,973.5	14,000,000
5	18,533.5	0.01	13,427,058.5	14,000,000
6	17,821.5 *	0.09	13,794,789.4	14,000,000
7	22,867.5	0.60	14,000,000	14,000,000

* Optimal value of MWLS.

larger ω_i for them (e.g., 10 in lieu of 3) to enforce the model to supply them. However, this would be at the cost of escalating the total load shedding level of the entire distribution network.

5) *DG Placement and Topology Reconfiguration*: This section is specifically designed to demonstrate the joint impacts of optimal DG placement and topology reconfiguration on the maximum allowable outage length and, hence, on serving higher levels of loads following an extreme event. The proposed cases are defined as follows:

- Case 1) *Optimal DG placement with reconfiguration*
- Case 2) *Random DG placement with reconfiguration*
- Case 3) *Optimal DG placement without reconfiguration*
- Case 4) *Random DG placement without reconfiguration*

In order to thoroughly reveal the significance of the optimal DG placement and topology reconfiguration, it is assumed that, at this time, the load priorities are the same and ω_i is 1 for all the buses. Furthermore, in the random DG placement cases, DG resources are considered to be arbitrarily installed at buses 5, 13, 21, 24, and 32.

Fig. 3 depicts the network's total length of lines that can be disrupted to maintain an equal value of load shedding, set by β . It is observed that when both the optimal DG placement and topology reconfiguration are considered, a greater length in km can be on outage to serve the same level of loads. For instance, the maximum load shedding level is 2,143.75 kW for $\beta = 0.25$, which means that by deploying the joint optimal DG placement and topology reconfiguration, 26.6 km can be damaged, whereas by placing the DGs randomly and without network reconfiguration, 17.2 km is permitted to be damaged in order to maintain the same 2,143.75 kW of load shedding.

C. Taiwanese 94-Bus Real Distribution Network

1) *Test System*: The second case study is a modified version of a distribution system of Taiwan Power Company

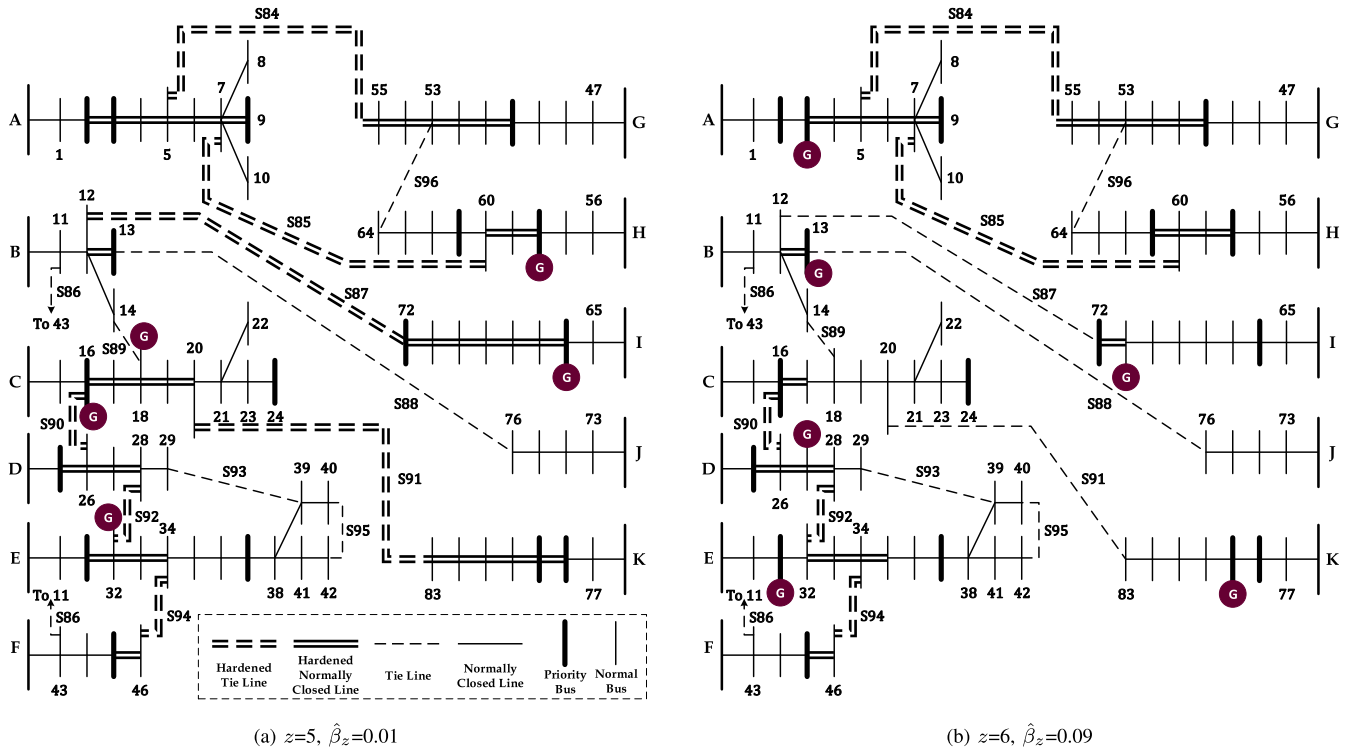


Fig. 4. Optimal planning schemes for the 94-bus distribution system.

(TPC) [42], with 11 feeders, 83 normally closed lines, and 13 tie lines. This is a 3-phase 11.4 kV system, with the total active and reactive demand of 28.35 MW and 20.77 MVar, respectively. Its voltage and line flow limits are assumed to be as the same as those in the 33-bus system. We have used the impedance data in [42] to calculate the length of distribution lines. While the line with the lowest resistance (i.e., line 63-64) is considered to be 100 m long, the length of the other lines increases proportional to their resistances. The system possesses backup DGs, each with a capacity of 2,000 kW, and the total planning budget is assumed \$14,000,000 for this network.

2) *Base Case*: Similar to the 33-bus test system, in the base case the entire capacity of DGs is used to pick up some loads. If, for example, six DGs are placed in the network, the base load shedding S_b will be $28,350 \text{ kW} - 6 \times 2,000 \text{ kW} = 16,350 \text{ kW}$, and its corresponding $\bar{\gamma}$ is 29.15 km.

3) *Conservativeness Degree Selection*: In order to find the best possible β and the optimal number of DGs, resulting in the minimum load shedding level, an LUT is similarly generated for the 94-bus test system. A shortened instance is given in Table V. It is revealed that the first possible row in the table comes with $n = 5$, having $\hat{\beta}_z = 0.09$ and $\hat{\zeta}_z = 17,821.5 \text{ kW}$. Repeating the process for the other values of z , a summary of results is given in Table VI, for $z = 4$ to $z = 7$. Observe that $z = 6$ contributes to the minimum load shedding value at 17,821.5 kW. Hence, six DGs are placed in the network, and its corresponding $\hat{\beta}_6$ is 0.09.

4) *Optimal Planning Scheme*: This section provides the optimal hardening schemes for the 94-bus test system. The planning results for five and six DGs are shown in Fig. 4. It

is worth mentioning that the weight of all the priority loads is three times larger than that of the other loads—similar to the 33-bus test system. In the 94-bus network, different buses have significantly different load values. For instance, a 2,000 kW demand at node 79 is inevitably powered for both $z = 5$ and $z = 6$, whereas priority buses 24 and 37, with respectively insignificant load levels of 50 and 20 kW, are not picked up. As mentioned in the 33-bus network, the topological locations of very small priority loads may not allow to invest a large amount of money in hardening kilometers of connecting distribution lines. If, however, these ignored priority loads must be supplied, one may specifically consider a large ω_i for their desired buses.

The utilization and hardening of tie lines as critical paths between DG-powered MGs and islands are also demonstrated in Fig. 4. It is observed that a total of seven and five tie lines respectively make contribution to flowing the power to the loads, for $z = 5$ and $z = 6$, and hence, these are hardened, as well, showing the significance of the topology reconfiguration.

The radial operation of multiple DGs within MGs can also be seen in Fig. 4(a). Indeed, the optimal planning scheme for $z = 5$ results in a total of three MGs in the 94-bus network. Although two of these MGs are energized through single generation units, the other maintains the radial operation while simultaneously committing three DGs at nodes 16, 18, and 32 thanks to the utilized radiality constraints expressed in Section II.

V. CONCLUSION

This paper presented a new strategy based on Information Gap Decision Theory for enhancing the distribution system

resilience. The outage status of distribution lines following a natural disaster was considered as an uncertain parameter. The proposed method deployed a robustness function to maximize the impacts of topological uncertainties associated with line outages, while ensuring that a minimum load level can be supplied. The model was formed considering several equality and inequality constraints using the DistFlow equations. The proposed MILP model was solved by the commercial solver CPLEX using the GAMS modeling language. The presented approach was applied to 33-bus and 94-bus distribution networks, and the provided results demonstrated that the optimal DG resource placement, topology reconfiguration, and optimal line hardening schemes can considerably boost the resilience in distribution networks. Additionally, the proposed conservativeness degree selection algorithm indicated that with a limited budget, a trade-off analysis between the costs of DG allocation and line hardening can contribute to the best possible conservativeness degree, based on the minimization of the entire network's weighted load shedding level.

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